Joint Optimization of Signal Quantization and Detection in Cell-Free MIMO: A Quantization Aware GNN Approach

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***Abstract*—Cell-Free Massive Multiple-Input Multiple-Output (MIMO) systems achieve superior spectral efficiency through distributed Access Point (AP) cooperation, but their practical deployment is severely constrained by limited fronthaul capacity. This paper proposes a Joint Quantization-Aware Training (QAT) framework that co-optimizes adaptive signal quantization at the APs and centralized detection at the Central Processing Unit (CPU). At the AP side, a triple-head Policy Network employs Gumbel-Softmax relaxation to dynamically select optimal quantization bit-widths from the set $\{0, 4, 8, 12, 16\}$ for received signals, channel estimates, and local demodulation results based on instantaneous channel conditions. At the CPU side, a Mean-Field Graph Neural Network (GNN) detector, integrated with a multi-head Transformer, robustly combines heterogeneous-precision signals for multi-user interference cancellation. The entire framework is trained end-to-end with a task-oriented loss that minimizes detection error subject to a fronthaul bit-budget constraint. Numerical evaluations, averaged over multiple large-scale fading realizations under the 3GPP TR 36.814 (UMi) model, demonstrate that the proposed QAT-GNN achieves Bit Error Rate (BER) performance comparable to full-precision baselines (e.g., MMSE-PIC) while reducing the fronthaul load by over $29\times$. The framework exhibits graceful degradation under link failures (maintaining connectivity at 50% AP dropout) and remarkable robustness to imperfect Channel State Information (CSI) and pilot contamination, significantly outperforming classical linear detectors.**

I. Introduction

The evolution toward the sixth generation (6G) of wireless networks is characterized by a fundamental paradigm shift from cell-centric architectures to user-centric Cell-Free Massive Multiple-Input Multiple-Output (MIMO) systems [1], [2]. By deploying a large number of distributed Access Points (APs) that cooperatively serve User Equipments (UEs) via a Central Processing Unit (CPU), Cell-Free MIMO offers significant theoretical gains in spectral efficiency and macro-diversity [3]. However, the practical realization is constrained by the fronthaul capacity bottleneck. Transmitting full-precision in-phase and quadrature (IQ) samples from massive AP arrays imposes an unsustainable load on limited-capacity fronthaul links [4], [5].

Traditionally, fronthaul compression relies on fixed-rate quantization methods, which are fundamentally agnostic to the downstream detection task [6]. Conventional schemes typically assign equal bit-widths to all APs regardless of their instantaneous Channel State Information (CSI), leading to inefficient resource utilization. For instance, an AP experiencing deep fading contributes little to the final detection but consumes the same fronthaul bandwidth as a strong link, thereby wasting valuable fronthaul capacity.

A. Related Work

1) Deep Learning for MIMO Detection: Deep learning (DL) has emerged as a powerful tool for physical layer optimization. Model-driven approaches, such as DetNet [7] and OAMP-Net2 [8], have been successfully applied to MIMO detection, demonstrating near-optimal performance with lower complexity than classical iterative methods. Recent works have explored Graph Neural Network (GNN)-enhanced detectors for decentralized MIMO systems [9], yet these primarily focus on the detection algorithm itself under ideal fronthaul assumptions, rather than the joint optimization with distributed quantization constraints.

2) Task-Oriented Communication: Task-Oriented Communication (TOC) leverages the Information Bottleneck (IB) principle to transmit only task-relevant features [10], [11]. While IB-based methods have been proposed for multi-device inference and semantic information recovery [12], there is a critical gap: existing frameworks often separate the resource allocation from the signal detection process or rely on static bit-allocation strategies that do not adapt to instantaneous CSI fluctuations.

B. Contributions

This paper proposes a Joint Quantization-Aware Training (QAT) framework for Cell-Free MIMO. Unlike prior works, we integrate a triple-head Resource Inference Module at each AP with a Mean-Field GNN at the CPU. The main contributions are summarized as follows:

- We propose a differentiable triple-head bit-selection mechanism utilizing Gumbel-Softmax relaxation, enabling end-to-end training of discrete quantization levels $\{0, 4, 8, 12, 16\}$ bits for heterogeneous signal components (received signals, channel estimates, and local demodulation results).
- We design a Mean-Field GNN detector that incorporates quantization precision as an attention bias. The architecture employs a Transformer encoder to robustly aggregate signals with heterogeneous quality, effectively ignoring silenced nodes.
- We provide extensive numerical results averaged over multiple large-scale fading realizations. We compare the proposed method with strong baselines, including linear

Minimum Mean Square Error with Parallel Interference Cancellation (MMSE-PIC). The system maintains robust performance even with 50% AP dropout and achieves detection comparable to full-precision baselines with approximately 96 bits per link, representing a 29× reduction in load compared to full 32-bit representations.

II. System Model and Problem Formulation

We consider the uplink of a Cell-Free Massive MIMO system with L distributed Access Points (APs) and K single-antenna User Equipments (UEs). Each AP is equipped with N antennas.

A. Physical Layer Channel Model

The channel vector between user k and AP l is denoted as $\mathbf{h}_{lk} = \sqrt{\beta_{lk}}\mathbf{g}_{lk} \in \mathbb{C}^{N\times 1}$, where β_{lk} represents the large-scale fading coefficient following the 3GPP TR 36.814 Urban Micro (UMi) path loss model with log-normal shadow fading, and $\mathbf{g}_{lk} \sim \mathcal{CN}(\mathbf{0}, \mathbf{R}_{lk})$ models the small-scale fast fading. Here, $\mathbf{R}_{lk}$ denotes the spatial correlation matrix at the AP array, reflecting practical dense antenna array deployments. The collective channel matrix at AP l from all K users is defined as $\mathbf{H}_l = [\mathbf{h}_{l1}, \ldots, \mathbf{h}_{lK}] \in \mathbb{C}^{N\times K}$. The received baseband signal at AP l is:

$$\mathbf{y}_l = \sum_{k=1}^{K} \sqrt{p}\mathbf{h}_{lk}s_k + \mathbf{n}_l = \mathbf{H}_l\mathbf{s} + \mathbf{n}_l, \tag{1}$$

where p is the uplink transmit power, $\mathbf{s} = [s_1, \ldots, s_K]^T \in \mathcal{X}^K$ is the transmitted QPSK symbol vector, and $\mathbf{n}_l \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I}_N)$ is the additive white Gaussian noise (AWGN) vector. In practical systems with imperfect CSI, the AP estimates the channel as $\hat{\mathbf{H}}_l = \mathbf{H}_l + \mathbf{E}_l$, where $\mathbf{E}_l$ represents the channel estimation error matrix with mutually independent entries drawn from $\mathcal{CN}(0, \tau_{est}^2)$.

B. Local Processing and Learnable Step Size Quantization (LSQ)

Due to the lack of global CSI and data at individual APs, computing the exact joint LMMSE receiver locally is infeasible. Therefore, each AP employs a local combiner based only on its local estimated channel $\hat{\mathbf{H}}_l$. Specifically, we use the local Maximal Ratio Combining (MRC) scheme to obtain an initial symbol estimate $\hat{\mathbf{s}}_l = \hat{\mathbf{H}}_l^H\mathbf{y}_l$.

To satisfy the stringent fronthaul capacity constraints, the APs must quantize the received signal $\mathbf{y}_l$, the estimated channel $\hat{\mathbf{H}}_l$, and the local estimate $\hat{\mathbf{s}}_l$ before forwarding them to the CPU. We employ a Learnable Step Size Quantizer (LSQ) $\mathcal{Q}(\mathbf{x}; b, s)$ for any feature tensor $\mathbf{x}$. For a chosen bit-width $b \in \mathcal{B}$, the available quantization levels are $\mathcal{V} = \{v \in \mathbb{Z} : -2^{b-1} \le v \le 2^{b-1} - 1\}$. The LSQ operation is defined as:

$$\mathcal{Q}(\mathbf{x}; b, s) = \text{round}\left(\text{clamp}\left(\frac{\mathbf{x}}{s}, v_{min}, v_{max}\right)\right) \cdot s, \tag{2}$$

where s is a strictly positive, learnable scale factor parameterizing the quantizer's dynamic range. Unlike traditional fixed uniform quantizers, s is optimized jointly with the neural network weights via backpropagation. Since the round$(\cdot)$ function is non-differentiable, we use the Straight-Through Estimator (STE) during training, implemented via the surrogate gradient: $\mathcal{Q}_{train}(\mathbf{x}) = \mathbf{x} + \text{stop_gradient}(\mathcal{Q}(\mathbf{x}) - \mathbf{x})$.

C. Optimization Problem Formulation

The Central Processing Unit (CPU) receives the quantized feature sets $\mathcal{F}_l = \{\mathbf{y}_l^q, \hat{\mathbf{H}}_l^q, \hat{\mathbf{s}}_l^q\}$ from all APs to reconstruct the user symbols $\mathbf{s}$. Let $b_{l,i} \in \mathcal{B} = \{0, 4, 8, 12, 16\}$ be the selected bit-width for component $i \in \{y, H, s\}$ at AP l. Note that $b = 0$ signifies that the component is entirely dropped. Our objective is to minimize the end-to-end detection Mean Squared Error (MSE) while strictly adhering to an average fronthaul bit budget C_{target}:

$$\min_{\Theta, \{b_{l,i}\}} \quad \mathcal{L}_{MSE} = \mathbb{E}\left[\left\|\mathbf{s} - \hat{\mathbf{s}}_{CPU}(\{\mathcal{F}_l\}_{l=1}^L; \Theta)\right\|^2\right] \tag{3}$$

$$\text{s.t.} \quad \frac{1}{L}\sum_{l=1}^{L}\sum_{i\in\{y,H,s\}} b_{l,i} \le C_{target}, \tag{4}$$

where Θ encompasses all learnable parameters of the detector and quantizers. This constrained optimization is relaxed into an unconstrained Lagrangian objective function for end-to-end learning:

$$\mathcal{L}_{total} = \mathcal{L}_{MSE} + \lambda \max\left(0, \frac{1}{L}\sum_{l=1}^{L}\sum_{i\in\{y,H,s\}} b_{l,i} - C_{target}\right)^2, \tag{5}$$

where λ is a tunable penalty factor that controls the trade-off between detection accuracy and fronthaul compression (Pareto front).

III. Proposed Joint Quantization and Detection Framework

A. AP-Side Triple-Head Policy Network

To enable task-oriented dynamic quantization, each AP l is equipped with a lightweight triple-head Policy Network, denoted as π_ϕ. Its role is to independently infer the optimal bit-widths $b_{l,i}$ for the three components $i \in \{y, H, s\}$.

The policy network is modeled as a 3-layer Multi-Layer Perceptron (MLP) with a hidden dimension of 64 and ReLU activation functions. The input is a locally extracted statistical feature vector $\mathbf{a}_l \in \mathbb{R}^3$, which comprises the local received signal power $\|\mathbf{y}_l\|^2/N$, the Frobenius norm of the estimated channel $\|\hat{\mathbf{H}}_l\|_F^2$, and the local empirical Signal-to-Interference-plus-Noise Ratio (SINR).

The output of each head is a probability logits vector for the categorical distribution over the available bit-widths $\mathcal{B} = \{0, 4, 8, 12, 16\}$. To facilitate end-to-end differentiable backpropagation through discrete categorical sampling, we utilize the Gumbel-Softmax relaxation. The continuous approximation of the categorical selection weight for the j-th bit-width option is given by:

$$w_{l,i}^{(j)} = \frac{\exp((\log \pi_\phi(\mathbf{a}_l)_{i,j} + g_{l,i,j})/\tau)}{\sum_{m=1}^{|\mathcal{B}|}\exp((\log \pi_\phi(\mathbf{a}_l)_{i,m} + g_{l,i,m})/\tau)}, \tag{6}$$

where $g_{l,i,j}$ are i.i.d. noise samples drawn from the standard Gumbel distribution Gumbel$(0,1)$, and $\tau > 0$ is the annealing temperature parameter. The effective differentiable bit-width utilized in the continuous LSQ penalty during training is computed as $b_{l,i} = \sum_{j=1}^{|\mathcal{B}|} w_{l,i}^{(j)} \mathcal{B}_j$.

B. CPU-Side Detection with Bit-Aware Attention

Upon receiving the quantized features, the CPU employs a hybrid architecture comprising a Mean-Field Graph Neural Network (GNN) for initial spatial interference cancellation, followed by a Transformer encoder for refined multi-user detection.

1) Mean-Field GNN Layer: The CF-MIMO network is modeled as a bipartite graph connecting APs and UEs. For a user node k at iteration t, the GNN aggregates spatial messages from all connected APs. Crucially, the bit-width $b_{l,y}$ is concatenated with the quantized physical features to inform the network of the varying fidelity of the messages:

$$\mathbf{m}_{lk}^{(t)} = \text{MLP}_{msg}\left([\mathbf{h}_{lk}^{q} \parallel \mathbf{y}_{l}^{q} \parallel b_{l,y}]\right), \tag{7}$$

$$\mathbf{z}_{k}^{(t+1)} = \text{MLP}_{update}\left(\mathbf{z}_{k}^{(t)}, \frac{1}{L}\sum_{l=1}^{L} \mathbf{m}_{lk}^{(t)}\right), \tag{8}$$

where $\parallel$ denotes concatenation, $\mathbf{z}_k^{(t)}$ is the hidden state of user k, and the aggregation function is implemented as a spatial MEAN pooling over L APs. Explicitly passing the bit-width enables the GNN to automatically down-weight noisy messages originating from heavily quantized or dropped ($b = 0$) AP links.

2) Bit-Aware Transformer: To capture complex inter-user interference patterns over the aggregated features $\mathbf{Z} \in \mathbb{R}^{K \times d_{model}}$, we deploy a Transformer encoder equipped with a novel *Bit-Aware Multi-Head Attention (BA-MHA)*. Standard attention relies solely on the feature content via query $\mathbf{Q} = \mathbf{Z}\mathbf{W}_Q$ and key $\mathbf{K} = \mathbf{Z}\mathbf{W}_K$, where $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{d_{model} \times d_k}$ are learnable linear projection matrices.

We introduce an explicit topological bias based on the overall quantization resolution of the contributing APs. Let $\mathbf{e}_l$ be the one-hot encoding of the bit-widths assigned to AP l. The modified attention coefficient from user k to an AP representation l is expressed as:

$$\alpha_{kl} = \frac{\exp\left(\frac{\mathbf{q}_k^T \mathbf{k}_l}{\sqrt{d_k}} + \text{MLP}_{bit}(\mathbf{e}_l)\right)}{\sum_{l'} \exp\left(\frac{\mathbf{q}_k^T \mathbf{k}_{l'}}{\sqrt{d_k}} + \text{MLP}_{bit}(\mathbf{e}_{l'})\right)}. \tag{9}$$

This architecture enforces a form of task-oriented spatial filtering, dynamically focusing the CPU's processing capacity on high-precision spatial streams.

C. Training and Implementation Details

To guarantee full reproducibility, we detail the end-to-end training strategy. The dataset is dynamically generated during training based on the 3GPP UMi channel model, simulating random large-scale fading topologies in a 1×1 km^2 area. We utilize the Adam optimizer across 200 total epochs, grouped into batches of size 256.

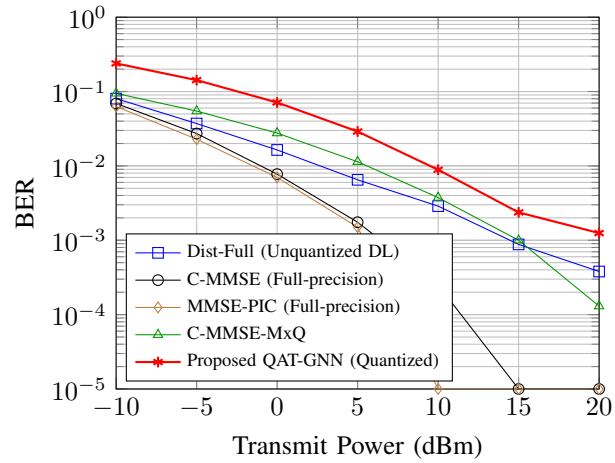

Fig. 1. BER performance under perfect CSI with $L = 16, K = 8$. All quantized schemes target approximately 96 bits per link.

The training is strictly divided into two phases. In Phase 1 (50 epochs), we train the hybrid GNN-Transformer detector under fixed high-resolution uniform quantization (learning rate 10^{-3}) to initialize a stable detection backbone. In Phase 2 (150 epochs), we activate the Policy Network and LSQ modules to jointly learn the dynamic bit allocation and detection (learning rate 5×10^{-4}). The Gumbel-Softmax temperature τ is linearly annealed from 1.0 down to 0.1 to encourage hard, discrete bit decisions. The Lagrangian penalty factor λ is tuned precisely between $[0.0, 0.2]$ to establish the Pareto frontier of the detection BER versus average fronthaul bit consumption.

IV. Numerical Results

Evaluations are performed in a 1×1 km^2 area with $L = 16$ Access Points (APs) and $K = 8$ User Equipments (UEs) (unless specified otherwise). The channel follows the 3GPP TR 36.814 (UMi) path-loss model with 8 dB shadow fading and Rayleigh fast fading. Each AP is equipped with $N = 4$ antennas. To address reproducibility and avoid scenario-specific bias, all results are averaged over 5 random large-scale fading seeds, ensuring geometry-independent evaluation.

A. Performance under Perfect and Imperfect CSI

To rigorously evaluate our approach against classical and deep learning state-of-the-art (SOTA), we compare the proposed QAT-GNN against: 1) Centralized MMSE (C-MMSE) with full-precision; 2) MMSE-PIC, a stronger non-linear interference cancellation baseline; 3) C-MMSE-MxQ, an SNR-adaptive quantized C-MMSE baseline; and 4) Dist-Full, an unquantized full-precision version of our deep learning model representing the performance upper bound of the proposed distributed architecture.

Fig. 1 shows the BER under perfect CSI. The proposed QAT-GNN (average 96 bits/link) significantly outperforms the improved C-MMSE-MxQ baseline at moderate to high SNR. While the unquantized, centralized C-MMSE and MMSE-PIC define the lower bounds with $\mathcal{O}((LN)^3)$ complexity, our proposed framework maintains a highly competitive BER with

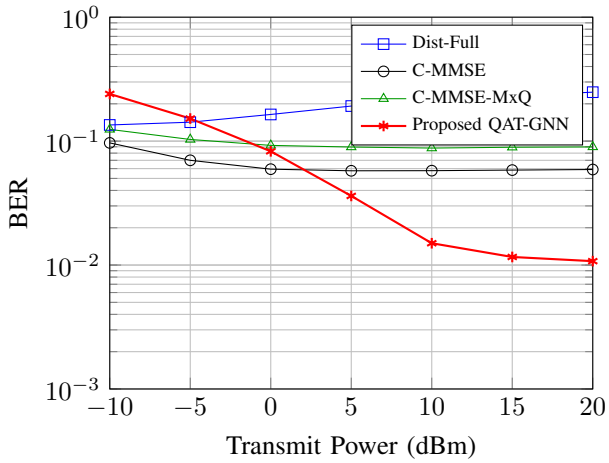

Fig. 2. BER performance under imperfect CSI ($\tau_{est} = 0.05$). The proposed method exhibits significant gain over linear baselines by learning to suppress estimation noise.

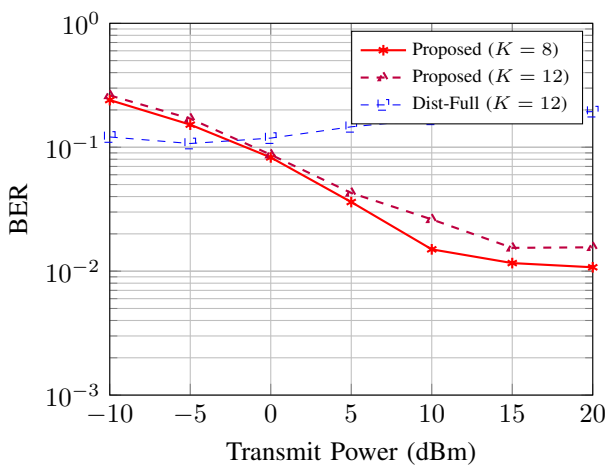

Fig. 3. Zero-shot generalization to $K = 12$ users using the model trained on $K = 8$ under imperfect CSI.

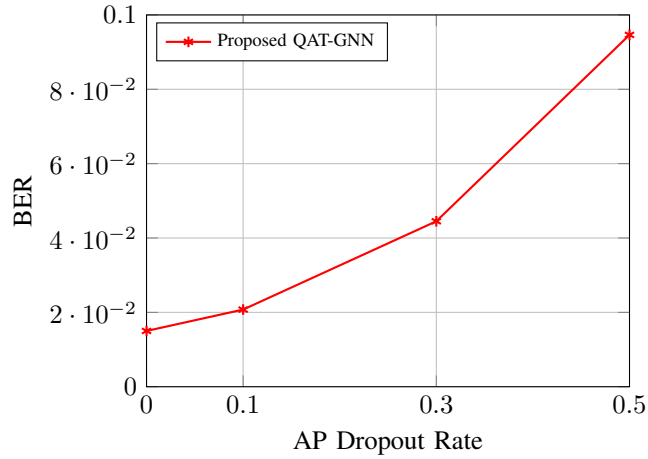

Fig. 4. BER degradation under varying AP Dropout Rates at $P_{tx} = 10$ dBm, demonstrating system robustness to link failures.

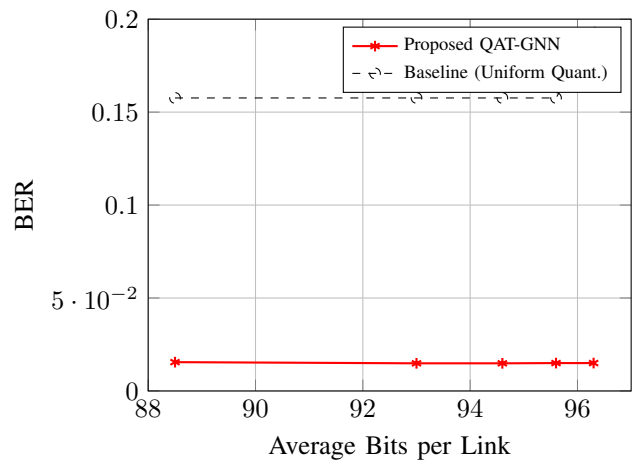

Fig. 5. Pareto front: Fronthaul bit-rate vs. BER at $P_{tx} = 10$ dBm, obtained by varying the penalty λ under imperfect CSI.

localized processing and severe fronthaul constraints (over $6\times$ compression compared to Dist-Full).

In Fig. 2, we evaluate imperfect CSI with error variance $\tau_{est} = 0.05$. This scenario accounts for practical pilot contamination and estimation noise. The proposed framework demonstrates superior robustness, achieving a BER of 0.015 at 10 dBm, significantly outperforming full-precision C-MMSE (0.057) and the quantized baseline C-MMSE-MxQ (0.087). Noticeably, even the unquantized Dist-Full baseline suffers heavily from channel errors, whereas the proposed QAT-GNN effectively learns to perform task-oriented filtering, masking out noisy AP estimates.

B. Generalization and AP Dropout Robustness

The GNN was trained on a fixed $K = 8$ setup. Thanks to the permutation-equivariant nature of the graph pooling and attention mechanisms, the model can natively handle varying user dimensions. Fig. 3 confirms this zero-shot generalization to $K = 12$ users.

Furthermore, in practical CF-MIMO networks, fronthaul links may completely fail. Fig. 4 evaluates the system under AP link failures (AP Dropout). While centralized linear methods would fail abruptly due to rank deficiency when losing too many antennas, our proposed GNN architecture gracefully degrades. Even at a severe 50% AP dropout rate, the system remains operational, yielding a BER of 0.094 at 10 dBm (averaged across 5 random topological seeds).

C. Pareto Frontier, Ablation, and Complexity

Fig. 5 illustrates the Pareto front for average bit-rate vs BER. By tuning the penalty factor λ during training, our model maintains robust performance (0.015 BER) even as the bit budget decreases to 88.5 bits, unlike the uniform quantization baseline which degrades rapidly to > 0.15 BER.

Fig. 6 provides an ablation study. It is observed that the Bit-Aware Attention is the most critical component, as its removal leads to significant performance loss (more than doubling the BER from 0.015 to 0.036 at 10 dBm).

Regarding end-to-end computational complexity, a standard full-precision centralized MMSE requires $\mathcal{O}((LN)^3)$ FLOPs

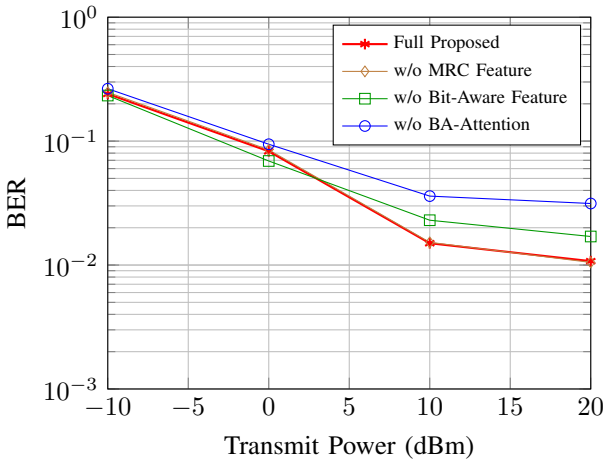

Fig. 6. Ablation study showing the impact of the MRC feature, bit-aware feature, and attention mechanism under imperfect CSI.

for matrix inversion. In contrast, the proposed system operates via distributed processing: the AP-side Policy Network requires merely ≈ 18k FLOPs per AP. At the CPU side, the graph aggregation and Transformer encoder scale at $\mathcal{O}(L^2 \cdot d + LKd)$ per layer. For large-scale Cell-Free systems where $L \gg K$, this linear-to-quadratic scaling in L with small hidden dimension $d = 128$ is far more computationally efficient than centralized cubic matrix inversions, establishing our approach as both communication- and computation-efficient.

V. Conclusion

This paper presented a Joint Quantization-Aware Training (QAT) framework for Cell-Free Massive MIMO systems. By integrating a triple-head policy network at the APs with a Mean-Field GNN-Transformer detector at the CPU, we achieved efficient task-oriented fronthaul compression (representing a 29× reduction in load) while maintaining high detection accuracy. Our numerical evaluations, averaged over diverse geometric scenarios, demonstrate that the proposed framework is remarkably robust to CSI errors, pilot contamination, and AP link failures, maintaining operational reliability in challenging environments where classical linear methods fail. This approach provides a scalable and robust solution for the deployment of 6G distributed architectures under practical fronthaul constraints.

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